

TITLE PAGE



Problem Statement ID –
26146



Problem Statement Title –
AI-Powered Monitoring & Analysis of
Bitcoin Transaction Traffic



Theme –
Blockchain & Cybersecurity



PS Category –
Software



Team ID –



Team Name (Registered on portal)
CryptoTrace AI



[Prototype](#)



[Github Repo](#)



[About](#)



AI-Powered Bitcoin Transaction Intelligence & Investigative Platform



Proposed Solution (Describe your Idea/Solution/Prototype)

CryptoTrace AI is an offline, AI-powered platform that ingests bulk Bitcoin transaction and network metadata, builds an interactive entity graph, detects suspicious activities using machine learning, and provides explainable, prioritized leads for investigators through an intuitive dashboard.



Detailed explanation of the proposed solution

The system collects and processes transaction & network data (CSV/JSON/XML), enriches it using GeoIP/ASN databases, and correlates IPs, wallets, and transactions. It applies AI/ML models (Isolation Forest, DBSCAN) to detect anomalies and clusters. Risk scores and explainable insights are generated and visualized on an investigator dashboard with graphs, timelines, and reports.



How it addresses the problem

It helps investigators trace the flow of illicit funds, uncover hidden relationships, detect unusual patterns, and prioritize high-risk entities efficiently—reducing manual effort, increasing accuracy, and accelerating actionable intelligence.



Innovation and uniqueness of the solution

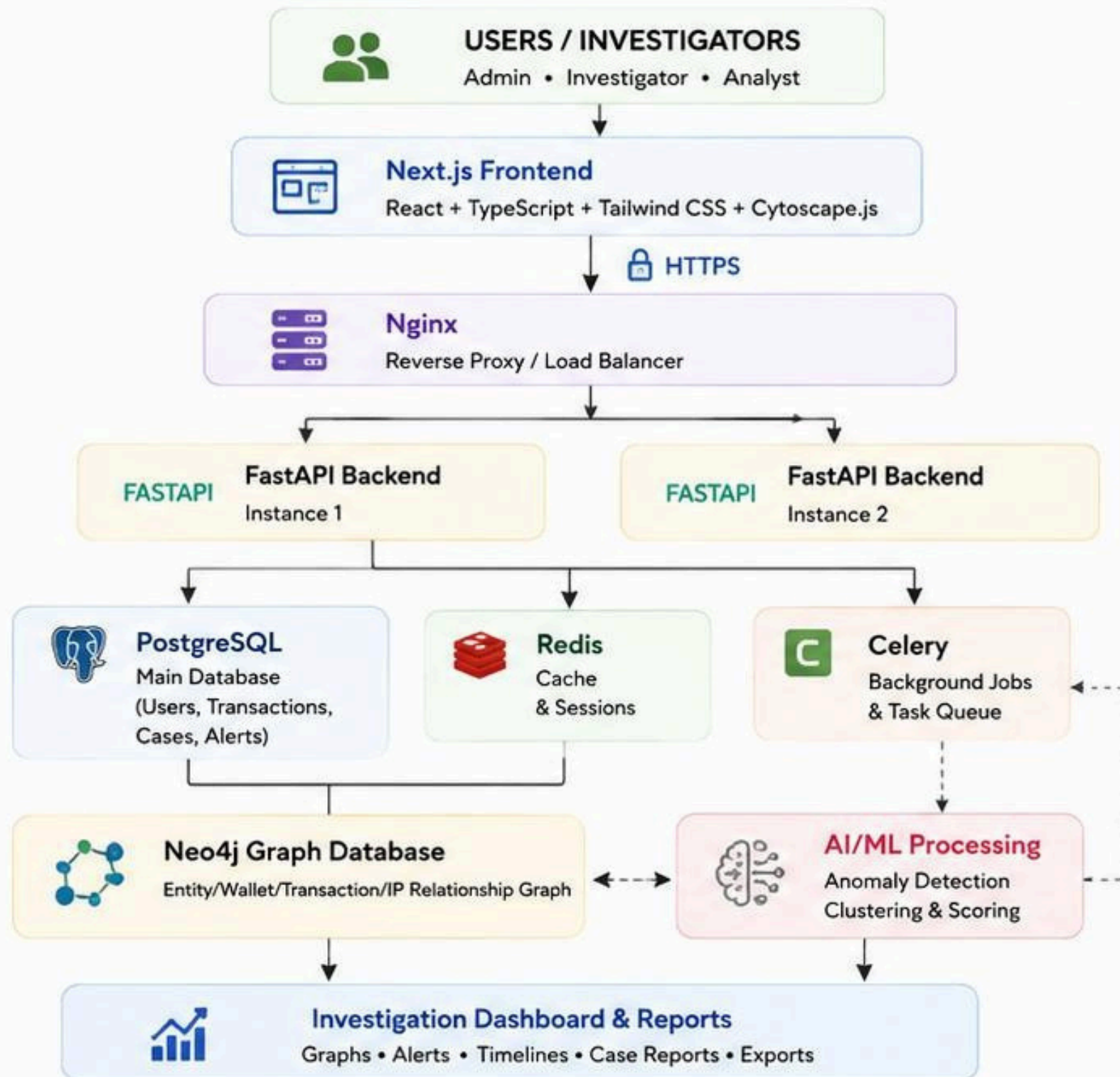
Combines network-layer metadata with blockchain data, AI-driven anomaly detection, explainable risk scoring, and interactive graph analysis—delivered as a fully offline, privacy-preserving platform for law enforcement and agencies.



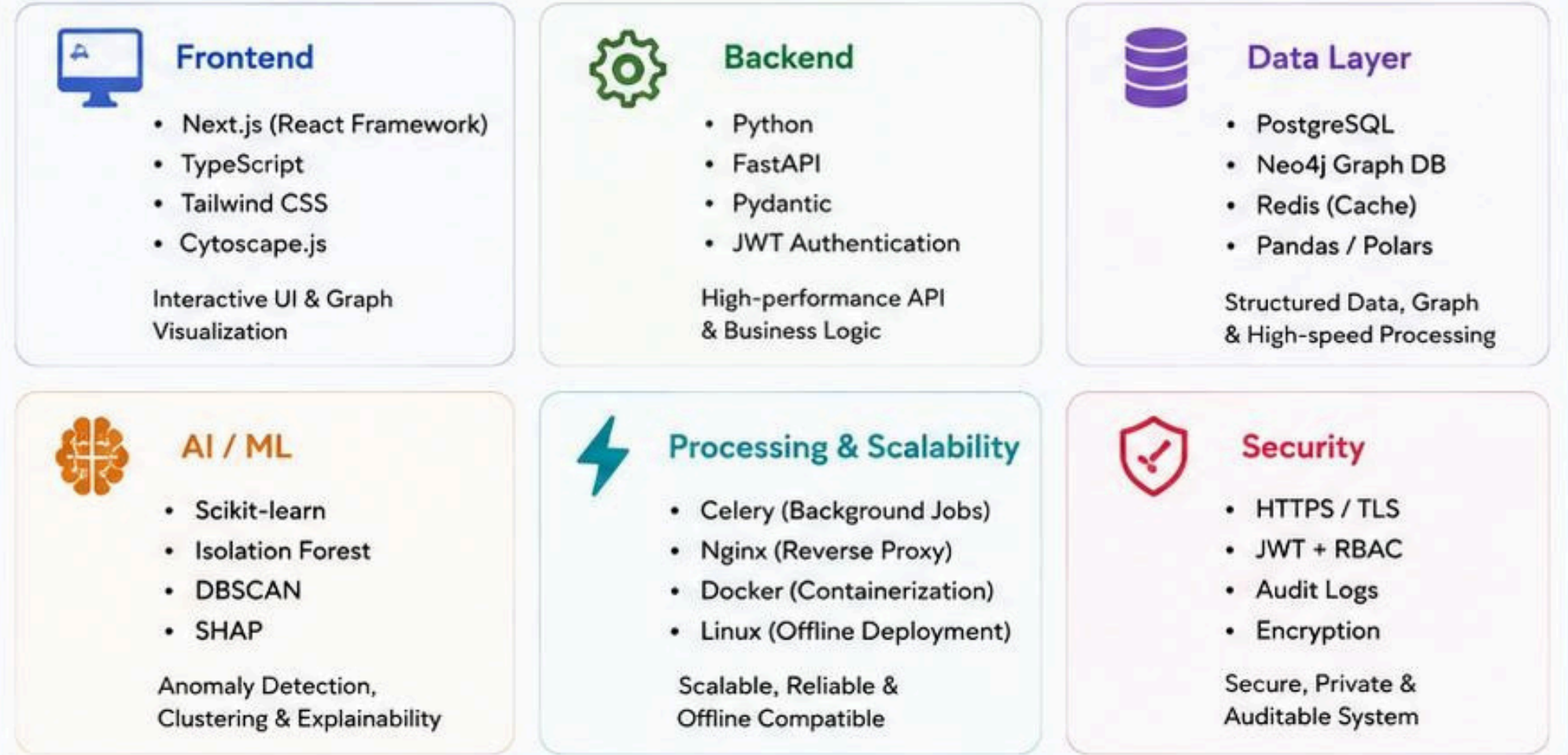
SOLUTION WORKFLOW



SYSTEM ARCHITECTURE



TECHNOLOGY STACK



KEY WORKFLOW



Goal: Detect suspicious cryptocurrency activities by correlating blockchain and network data using AI/ML and graph analytics, and deliver explainable, prioritized investigative leads.



Offline First



Explainable AI



Scalable & Secure

✓ Feasibility	⚠ Challenges / Risks	💡 Mitigation Strategy
 Uses open-source technologies such as Python, PostgreSQL, NetworkX and Scikit-learn.	 Large datasets may cause slow processing.	 Chunked ingestion + background workers + indexing + caching.
 Works with the synthetic dataset specified by the problem statement.	 IP-to-wallet correlation may be incomplete or noisy.	 Treat network correlation as supporting evidence, not definitive identity proof.
 Core system can operate offline on Linux.	 ML may produce false positives.	 Combine ML anomaly scores with graph/transaction features and human review.
 Graph-based analysis is suitable for tracing transaction relationships.	 Very large graphs can become difficult to visualize.	 Progressive graph expansion and filtering by time/risk/depth.
 Modular architecture allows future scaling.	 Maintaining explainability of AI decisions.	 SHAP + visible evidence/features behind every alert.
 Uses containerized services for reproducibility.	 Sensitive investigative data requires strong security.	 RBAC, encryption, audit logs and isolated offline deployment.




The system assists investigators by prioritizing suspicious activity and connecting evidence; it does not claim to identify a real-world person solely from a wallet or IP address.

Potential Impact on the Target Audience



Law Enforcement Agencies

Faster identification of suspicious wallets and transactions, stronger evidence collection, and better decision-making.



Financial Institutions & Exchanges

Enhanced AML compliance, reduced risk exposure, and improved monitoring of high-risk transactions.



Investigators & Analysts

Intuitive tools and visualizations reduce manual effort and speed up complex investigation workflows.



Policy Makers & Regulators

Data-driven insights and analytics help in framing better policies and understanding illicit trends.



Citizens & Society

Safer financial ecosystem with reduced cybercrime, fraud, ransomware and money laundering activities.

Benefits of the Solution



Social Benefits

Helps build a safer digital society by tackling financial crimes, protecting victims and increasing trust in financial systems.



Economic Benefits

Reduces financial losses due to fraud, ransomware and money laundering. Helps institutions save compliance and investigation costs.



Operational Benefits

Automates data analysis, reduces investigation time, improves accuracy and enables efficient allocation of investigative resources.



Environmental Benefits

Digital and paperless investigations reduce paper usage and operational overhead.



Legal & Regulatory Benefits

Supports regulatory compliance (AML/CFT), strengthens transparency and aids in legal proceedings.



Technological Benefits

Promotes the adoption of AI, Graph Analytics and Data Science for solving real-world challenges in law enforcement.



CryptoTrace AI empowers investigators with intelligent insights, accelerates detection of illicit activities, and contributes to a secure, transparent and responsible digital economy.



KEY RESEARCH PAPERS



Bitcoin: A Peer-to-Peer Electronic Cash System (2008)

Satoshi Nakamoto

The foundational whitepaper that introduced Bitcoin, its transaction model, timestamping, proof-of-work and peer-to-peer network.

bitcoin.org/bitcoin-paper



Detecting Anomalous Cryptocurrency Transactions

Pocher et al., 2023

Uses graph representation and ML (GCN & GAT) techniques to detect anomalous / illicit transactions for AML/CFT applications.

[Springer – Electronic Markets](#)



Bitcoin Address Clustering Using Multiple Heuristics

He et al., 2022

Clusters Bitcoin addresses into entities using heuristics like common-input ownership, change detection and Louvain algorithm.

[IET Blockchain – Wiley](#)



Deanonymisation of Clients in Bitcoin P2P Network

Biryukov et al., 2014

Shows how IP addresses can be linked to Bitcoin transactions through network-layer observations and timing.

[arXiv: 1405.7418](#)



Deanonymization in the Bitcoin P2P Network

Fanti & Viswanath, 2017

Analyzes transaction propagation and shows the risk of linking IP addresses with Bitcoin pseudonyms.

[NeurIPS 2017](#)



Graph Transformer for Illicit Bitcoin Transaction Detection

Lin et al., 2026

Proposes a wavelet-temporal graph transformer approach achieving high accuracy in detecting illicit transactions.

[Scientific Reports \(2026\)](#)

DATASETS USED

01 Elliptic Bitcoin Transaction Dataset

- 203,769 transaction nodes
- 234,355 edges
- Labeled as licit, illicit, unknown
- 166 features per node
- 49 temporal snapshots

[Kaggle – Elliptic Data Set](#)

02 BitcoinHeist Ransomware Dataset

- Jan 2009 – Dec 2018 data
- Address-level behavioral features
- Ransomware + Normal addresses
- ~2.9M instances, 10 features

[Kaggle – BitcoinHeist Dataset](#)

03 Bitcoin Blockchain Historical Data

- Raw blockchain data (blocks, transactions)
- Linked with Google BigQuery
- Ideal for building ingestion pipelines & graphs

[Kaggle – Bitcoin Blockchain](#)

04 Bitcoin Transactions 2024 Dataset

- 10,000 transaction records
- Fields: timestamp, block, hash, fee, input/output counts, etc.
- Great for testing & feature engineering

[Kaggle – Bitcoin TXs 2024](#)

05 Augmented Elliptic Dataset

- Addresses class imbalance in original Elliptic dataset
- Improves model performance and fairness
- Better for training ML models

[Kaggle – Augmented Elliptic](#)

TOOLS & REFERENCES



Python, Pandas, NumPy, Scikit-learn
Data processing & Machine Learning



NetworkX, PyG, Cytoscape
Graph construction & visualization



PostgreSQL
Data storage & indexing



GeoIP / ASN DB (MaxMind)
IP enrichment with Geo-location & ASN



SHAP
Explainable AI & Feature importance



Bitcoin.org, arXiv, IEEE, Springer, NeurIPS
Research papers & Technical references



These research papers and datasets provide the scientific foundation and real-world data required to build a robust, explainable and accurate Bitcoin transaction intelligence platform.